`model_simpleswing`: specification

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This documents the standard uniform swing model as implemented in `scripts/model/live/R/model_swing.R`. It is one of the two baselines the precinct × vote-mode model is compared against. Sections follow the same order as the main model's specification so the two can be read side by side.

1 Outcomes

The model works on precinct × vote-mode cells, with $m \in \{\text{mail}, \text{early}, \text{eday}\}$, and estimates two things per cell:

- a **turnout** correction, on the log scale, relative to a fixed baseline;
- a **two-party share** correction — the *swing* — on the raw share scale, relative to the precinct's prior lean.

The two are estimated by separate, independent regressions. Third-party share is not estimated at all: it is held fixed at the precinct's prior value $tp_i = \min(\max(\texttt{oth_base}_i, 0), 0.25)$.

2 The baseline

Both corrections are expressed relative to a baseline expectation of votes per cell, e_{im}. The precinct's expected total $\texttt{est_turnout}_i$ is split across modes using the precinct's own 2022 mode history, shrunk toward its county's split:

$$q_{im} = \frac{v_{im} + k\, c_{c(i)m}}{\sum_{m'} v_{im'} + k}, \qquad p_{im} = \frac{q_{im}}{\sum_{m'} q_{im'}}, \qquad e_{im} = \max\left(\texttt{est_turnout}_i \cdot p_{im},\ 0.5\right), \tag{1}$$

with $k = 40$, v_{im} the precinct's 2022 voters in mode m, and c_{cm} the county's mode share. This is the same baseline `model_cory` is given, and the same shrinkage constant the main model uses for its mode priors, so the three models start from a common point.

3 Turnout

Write c_{im} for the ballots counted so far in a cell and f_{im} for the fraction of the cell that has reported. The turnout outcome is the log ratio of counted to expected,

$$\ell_{im} = \log \frac{\max(c_{im},\, 0.5)}{e_{im}}, \tag{2}$$

fitted by two-way fixed effects on county and mode,

$$\ell_{im} = \alpha_{c(i)} + \gamma_m + \varepsilon_{im}, \tag{3}$$

weighted by e_{im} and estimated with `fixest::feols`. Only *counted-out* cells enter the fit: those with $f_{im} \geq 1$ and at least one two-party vote. Partially-counted cells contribute nothing to the turnout estimate.

Cells whose county or mode carries no fitted value fall back to the weighted mean for that mode, and then to the overall weighted mean.

The predicted cell total floors the fit at what is already counted:

$$\tau_{im} = \begin{cases} c_{im} & \text{if the cell is counted out,} \\ \max\big(c_{im},\ e_{im}\exp(\hat{\ell}_{im})\big) & \text{otherwise.} \end{cases} \tag{4}$$

Note what equations (3) and (4) together imply. The fitted correction $\hat{\ell}_{im}$ varies only at the county × mode level, so *all* precinct-level variation in predicted turnout comes from the baseline e_{im}; the fit supplies a shared multiplicative adjustment and nothing else. The model therefore never makes a precinct-specific turnout prediction.

4 Share

The share outcome is the swing away from the precinct's prior two-party lean $\texttt{dem2}_i$, on the raw share scale rather than the logit scale:

$$y_{im} = \frac{c_{im}^{\text{dem}}}{\max(c_{im}^{\text{dem}} + c_{im}^{\text{rep}},\ 1)} - \texttt{dem2}_i, \tag{5}$$

fitted by the same two-way specification (3) and the same weights. Here the fit set is wider: *any* cell with at least one two-party vote counts, whether or not it has finished reporting. The predicted share is clamped, $s_{im} = \min(\max(\texttt{dem2}_i + \hat{y}_{im},\ 0.001),\ 0.999)$.

5 Partial counts

Counted votes are kept exactly and the remainder is allocated at the predicted share:

$$r_{im} = (\tau_{im} - c_{im})(1 - tp_i), \qquad \text{dem}_{im} = c_{im}^{\text{dem}} + r_{im}\, s_{im}, \qquad \text{rep}_{im} = c_{im}^{\text{rep}} + r_{im}(1 - s_{im}). \tag{6}$$

A counted-out cell is substituted exactly, bypassing the prediction entirely.

6 Uncertainty

The interval comes from $B = 100$ **county bootstrap** replicates. Each replicate resamples the reporting counties with replacement and refits both equations with weights $w_{im} = e_{im} \cdot \#\{c(i)\}$, where $\#\{c\}$ is the county's multiplicity in that draw. The resulting statewide quantities are then re-centered on the point fit:

$$\tilde{S}^{(b)} = S^{(b)} - \overline{S} + \hat{S}_{\text{point}}. \tag{7}$$

Checkpoints at which nothing has been counted are skipped: with no reported cells the model has nothing to estimate.

Limitations

Stated plainly, and without prejudice to the model's purpose — it is a deliberately simple baseline and several of these are the price of that simplicity.

- **No ceiling.** Nothing bounds τ_{im} by the precinct's registrants. In practice this is far less dangerous than it sounds: because the fitted correction is constant within a county $\times$ mode cell, the model has very little variance available to blow up with. It cannot produce the kind of precinct-level overshoot an unconstrained per-precinct prediction can.
- **No adding up.** The three modes of a precinct are predicted independently and summed; nothing forces $\sum_m \tau_{im}$ to respect a precinct total. The per-cell floor in (4) is consistent with this, since there is no adding-up property for it to break.
- **Turnout and share are independent.** They are two unrelated fits, and turnout reaches the share aggregate only through the size of the remainder. Their correlation is not propagated. Measured within-mode $\rho(\tau, s)$ varies in sign across modes, so this is not a wash in either direction.
- **Asymmetric information use.** The share equation uses partially-counted cells; the turnout equation uses only counted-out ones. Early in the night the turnout fit can therefore be resting on very few cells.
- **The interval is a spread, not a posterior.** Re-centering (7) means the bootstrap cannot express bias, only dispersion, and that dispersion reflects between-county resampling alone — not residual precinct variance and not baseline error. Coverage comparisons against a model that draws from a posterior are not like-for-like, and should be read with interval *width* alongside.
- **Third party is not modeled.** It is a fixed per-precinct constant, so no uncertainty in it reaches the aggregate.